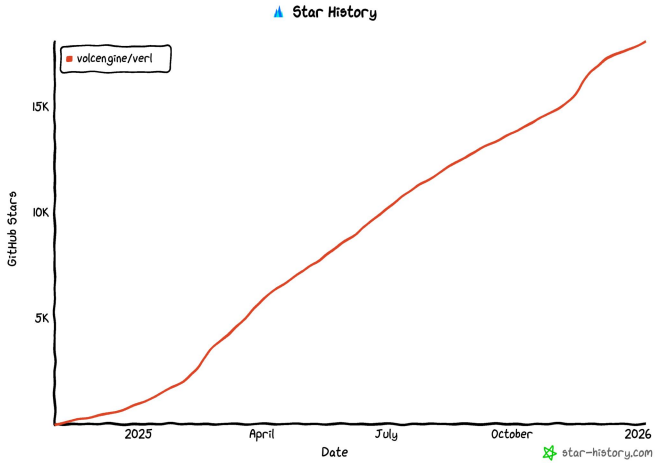
verl 0.7 release and roadmap

Xibin Wu
01/10/2026

verl's Open-Source Community



So far, verl has gained:

- 18k+ stars
- 3k+ forks
- 1.9k+ commits
- 490 contributors
- 2k+ issues

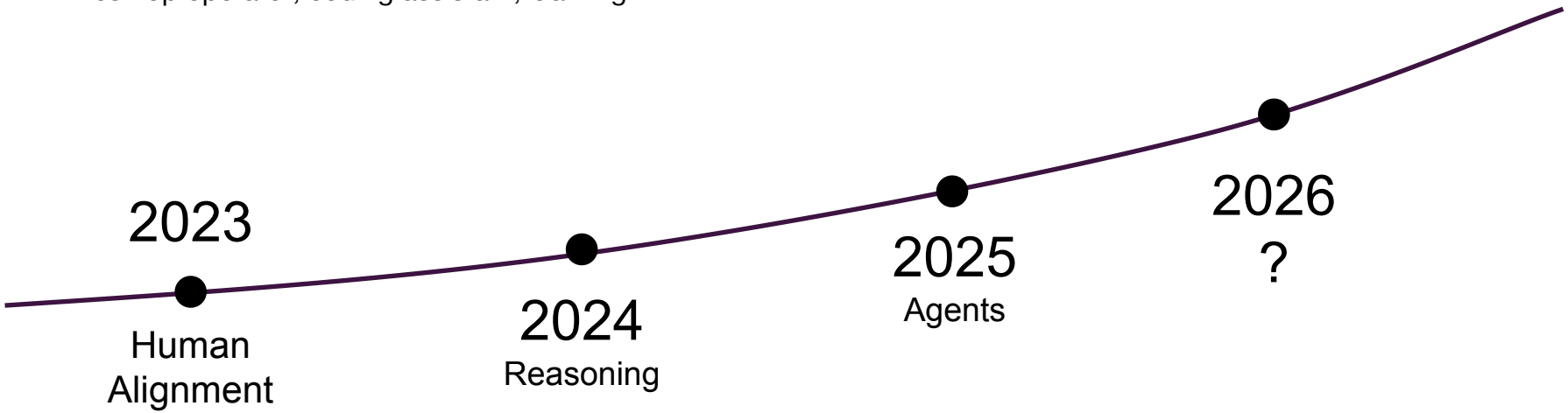
Many popular RL projects built on top of verl:

- TinyZero (12k stars)
- Easy-R1 (4.4k stars)
- Search-R1 (3.8k stars)
- SimpleRL-Zoo (3.8k stars)
- OpenManus-RL (3.8k stars)
- SkyThought (3.4k stars)
- …

RL trends over the past few years

Reinforcement learning (RL)

- Human Alignment with RLHF
- Reasoning: O1/Claude-3.7 performance on math benchmarks
- Image/video/music generation
- Agentic LLM tool using
- Desktop operator, coding assistant, Gaming…



RL trends 2026

Scaling law is still working for both Pretrain and RL

- Model size: MoE sparsity, 235B -> 671B -> 1T -> ?
- Compute: pretrain vs post-train: 0.9:0.1->0.7:0.3->?; pushing RL training to pretrain scale?
- Dataset: larger mixed tasks, lager sandbox cluster: SWE, GUI, Mobile, Game, etc

Agentic RL

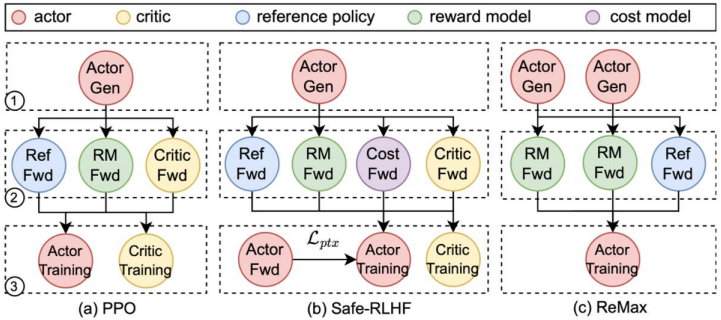
- From single-turn to multi-turn, from text reasoning to mutimodal
- Longer context: Qwen3 128K, GPT-5 400K, Gemini 3 Pro/Claude Sonnet 4.5 1M
- Byond context length: Claude skill, Context summary, Multi-Agent?

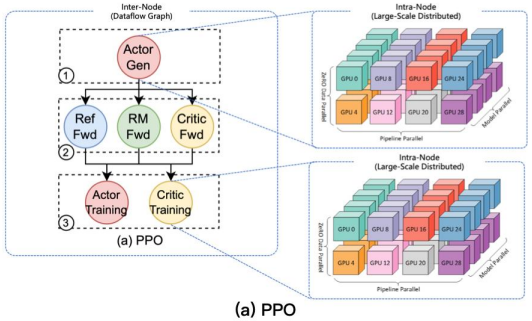
World Model and Embodied AI

- Native Omni Models: BAGEL, Qwen3-omni
- Robotics VLA

Architecture Review

HybridFlow = Single-controller(MPMD) + Multi-controller(SPMD)





Lession learned over the past year

- **Training**: lack of abstraction, redundant code for different backends.
- **Rollout**: spmd mode is intrusive and unfriendly to multi-turn conversation.
- **Single-controller**: coupled control flow and data flow, limiting scalability
- Lack native support for **asynchronous training**

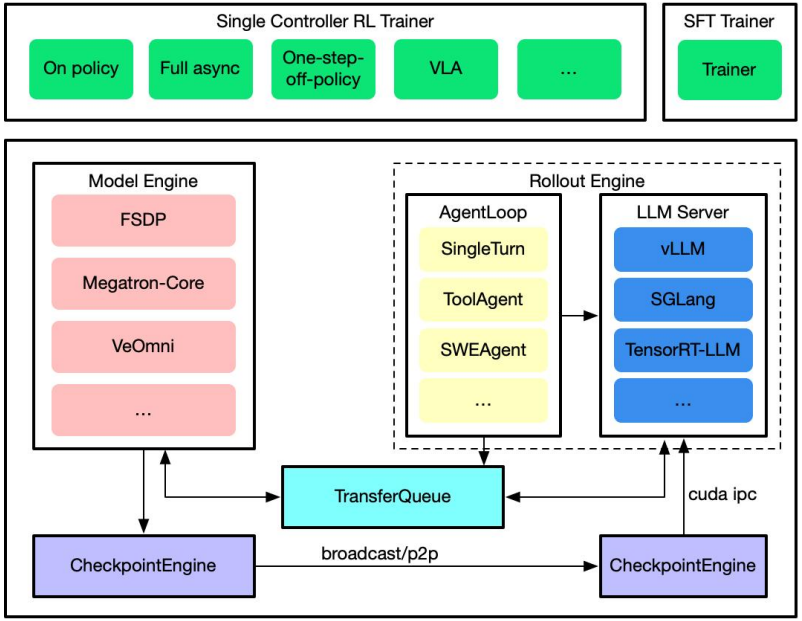
verl architecture in v0.7

verl-core: building blocks for RL pipline

- **Model Engine**: efficient training
- **Rollout Engine**: generation, environment interaction, reward calculation
- **TransferQueue**: data transmission, replay buffer
- **Checkpoint Engine**: weight synchronization

verl-trainer: RL pipelines built on top verl-core

- On policy: synchronous
- One-step-off/Fully async: asynchronous
- VLA
- ...



Model Engine

Core training engine for both **SFT** and **RL** training.

Goal: support larger model, longer context, pushing MFU to its limit.

API Design:

- Abstract Tinker-like API
- Run in SPMD, parallelelsm aware dispatch&collect
- Trainer agnostic to backend, new backend plugin without trainer code change

Feature:

- Multiple backends and various parallelism support
- Sequence balancing
- LoRA
- Efficient kernels for training
  - ✓ FlashAttenion
  - ✓ Liger-kernel
  - ✓ GroupGEMM/FusedMoE/DeepEP
  - ✓ FP8 training

```python
class BaseEngine:
    def initialize(self):
        """Instantiate or load the model, optimizer, and learning rate scheduler."""
        raise NotImplementedError

    def optimizer_zero_grad(self):
        """Zero the gradients of the optimizer."""
        raise NotImplementedError

    def optimizer_step(self):
        """Perform an optimization step using the optimizer."""
        raise NotImplementedError

    def lr_scheduler_step(self):
        """Advance the learning rate scheduler by one step."""
        raise NotImplementedError

    def forward_backward_batch(self, data: TensorDict, loss_function: Callable, forward_only=False) -> Any:
        """Perform a forward pass and optionally a backward pass on a batch of data."""
        raise NotImplementedError

    def get_per_tensor_param(self) -> tuple[Generator[tuple[str, torch.Tensor], None, None], Optional[dict]]:
        """Get a generator that yields per-tensor parameters and optional peft config."""
        raise NotImplementedError

    def to(self, device: str, model: bool = True, optimizer: bool = True, grad: bool = True):
        """Move model parameters, optimizer states, or both to the specified device."""
        raise NotImplementedError
```

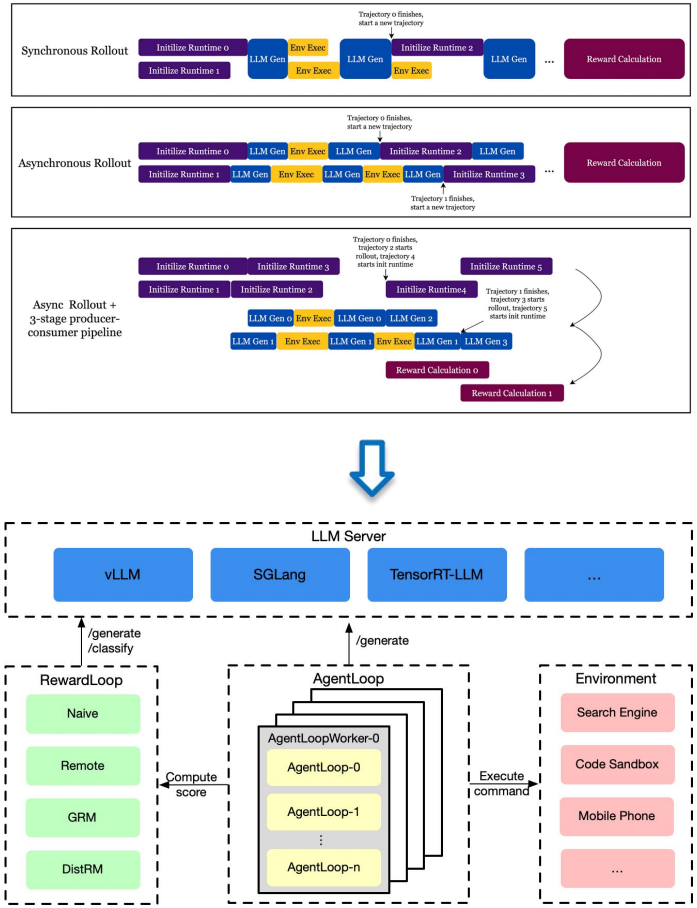
Backend	Parallelism	Performance	Support Model	New Model Days
FSDP	FSDP+SP	Dense medium/MoE Low	All transformer models	Day 0
MCore	DP+TP+PP +EP+CP	High	see Megatron-Bridge support list	few weeks or month
VeOmni	FSDP+SP+ EP	Medium	see VeOmni support list	~1 week

Rollout Engine

Agent loop centric multi-turn conversation rollout

- **LLM Server**: native inference server without intrusive modifications
  - ✓Weight synchronization
  - ✓FP8 online quantization
  - ✓Router replay
- **AgentLoop**: customizable agentic task loops, ReAct, SWE, GUI, etc
- **RewardLoop**: asynchronous reward calculation, support rule based, model based(GRM, DistRM)

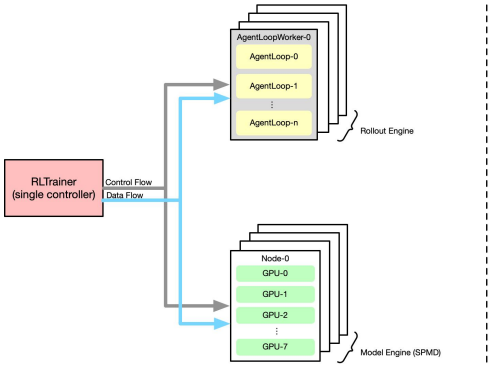
Source: https://novasky-ai.notion.site/skyrl-v0



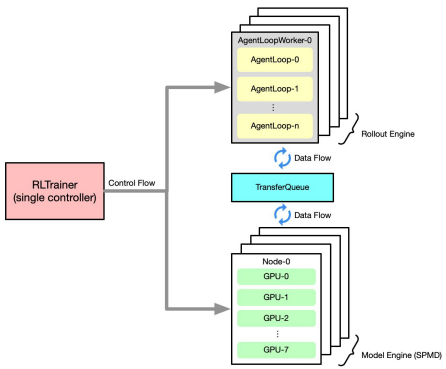
TransferQueue

Databus for all components, replay buffer.

- **Current limitation**: single controller handles both control and data flow, performance issue in large scale
- **Earlier fail attempt**: ray object store, high tensor serialization cost, lack fine-grained access, opaque gc mechanism
- **TransferQueue**
  - ✓ Zero-serialization
  - ✓ Extensible: multiple transport layer, TCP, RDMA
  - ✓ Fine-grained access: read/write/append subset columns
  - ✓ Proactive lifecycle management



(a) coupled data flow and control flow



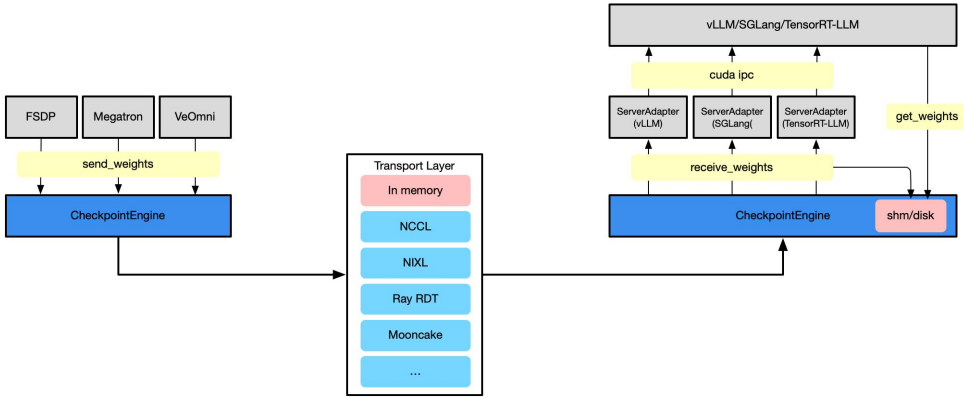
(b) decoupled data flow and control flow

Checkpoint Engine

Longer context and multi-turn agentic tasks amplify the "**long-tail**" problem, increase the need for **asynchronous training**.

Abstraction layer to synchronize weights between training and inference backends.

- **Unified API**: send_weights/receive_weights/get_weights
- **Extensible**: plugable transport backend
  - ✓ collective: nccl, hccl, uccl
  - ✓ p2p: nixl, mooncake
  - ✓ local cache: share memory, local disk



Backend	Topology	Performance	Elastic	Use case
Collective	all_gather+ broadcast	Very High	Low: rebuild nccl group	Fixed cluster
P2P	all_gather+ ring p2p	Medium/High	High: dynamic adjust	• Elastic rollout • Fault tolerance • Heterogeneous

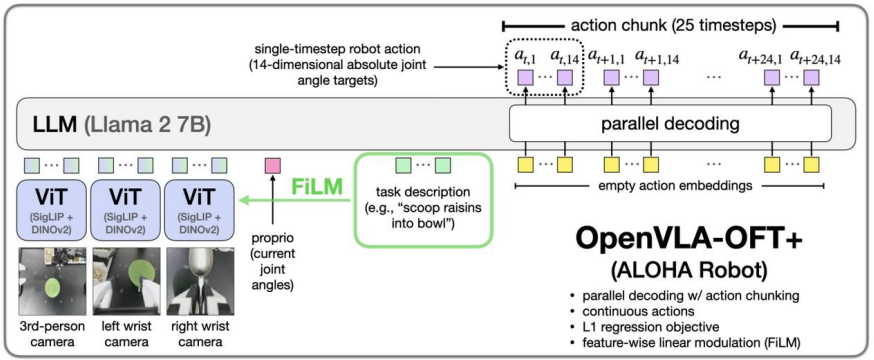
verl-trainer

Built on top verl-core, construct RL training pipelines flexibly.

- On policy trainer
- One-step-off-policy trainer
- Fully async trainer
- VLA trainer
- Many more custom trainer in [verl-recipe]



Source: Meituan Search AI Infra Team



Source: [openvla/openvla]

Roadmap: v0.8

Model Engine

- Deprecate DataProto by zero padded Tensordict
- Switch default to new model engine, mark legacy engine as deprecated
- Feature parity new and legacy model engine: LoRA/PEFT, etc
- Polish VeOmni engine to production ready
- Support MTP RL training
- Perfomance optimization
  - MoE multi-modal model training
  - Long context training: fine-grained activation recompuation/offload
- New model support: DeepSeek V3.2, etc

Rollout Engine

- New rollout engine: TensorRT-LLM
- Separate vllm worker from trainer, sync by cuda ipc

TransferQueue

- Merge TransferQueue recipe into main
- Optimize e2e image/video vlm training
- Optimize router replay transmission by TransferQueue

Checkpoint Engine

- Add checkpoint engine abstract interface
- Add NCCL, NIXL transport backend and more

Trainer

- Online policy distillation

Roadmap: v0.9

Trainer

- Merge Full async into main: refactor with verl-core component
- Elastic rollout in fully async

Model Engine

- Remove legacy model engine (fsdp_workers.py, megatron_workers.py)
- Support omni-model RL training: Qwen3-Omni, BAGEL

Rollout Engine

- New rollout engine: vllm-omni
- Speculative decoding

More agentic training recipe

- SWEAgent: WIP
- GUIAgent: TBD

Misc

- use **uv** for environment isolation

Thanks